

An Axiomatic Model II: Axiomatic Derivation and Interpretation of the (5, 1) Signature of the Wheeler-DeWitt Equation and the DeWitt Metric

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Abstract: Four axioms (orthogonality, atomicity, minimal cost, causality preservation) and one proposition define a discrete computational system. Because no free parameters enter the deduction, fitting is excluded; because the axioms close the search space, numerological tuning is blocked. The present paper is the second in the Axiomatic Model series and proceeds on the same axiom set as Paper I [5]. The complete axiomatic system comprises 15 axioms [6]; the present paper produces the following on *the same 4 axioms and 1 proposition* as Paper I, plus two registered statements of the complete system (listed exhaustively in Section 2.3). The four constituents of the Wheeler-DeWitt equation $\hat{H}\Psi[h_{ij}] = 0$ force its domain bit reading uniquely to (observer, superposition, time, space) = (0, 1, 0, 1) under the constituent correspondence rule of Definition 3 (Theorem 1), and the remaining 15 bit combinations are exhaustively excluded (Appendix A). The absence of time in the equation is a level-set projection produced by now-fixing plus rearrangement (Theorem 2), and the right-hand-side zero translates only as $\delta = 0$ (unfired), by positive-definiteness of the two-bracket norm (Theorem 3). On this foundation, the number of independent components of the DeWitt metric G_{ijkl} , $6 = T(3) = 7 - 1$, and its signature (5, 1) are cross-derived along two routes that draw their discriminating power from different bits of Theorem 1: on the time route, of the two times of the Banya equation (system time, domain time) the domain time is cut by now-fixing, yielding (5, 1); on the cost route, (5, 2) – observer = (5, 1). In the former picture, the (3, 1) of observable spacetime is the complementary case in which the system time is cut, and the two famous Lorentzian signatures are organized as the two cuts of “which time is fixed to now.” The latter extends to the universal structure of observation: an observer cannot place itself on its own observation list, so the signature of the observable world is always (5, 1). The remaining negative axis is identified as superposition (cost basis) and as the trace of system time (structural basis), and the standard convention of using the scale factor as internal time in minisuperspace is explained as a necessary consequence under the structural-basis reading – the present axiomatic system supplies an axiom-level ground for the sign structure of the DeWitt metric, for which no such ground had been offered since 1967, and the conformal-mode instability is reinterpreted not as a defect but as the structural scar of observer excision. As an independent check, we show that igniting the two axes switched off in the reading of Theorem 1 (observer, time) overlaps one-to-one with the two standard workarounds of the literature – Page-Wootters conditional probability and the thermodynamic arrow of time. As auxiliary results we present an axiom-number closed form $\sqrt{3}\ln 3/8$ approaching the loop-quantum-gravity free parameter, the Barbero-Immirzi γ , within 0.14%, together with one downstream item (the area gap) and one constituent-count reinterpretation (QNM); and as a forward-testable item the Hubble tension $H_0^{\text{local}}/H_0^{\text{CMB}} = \sqrt{57/50}$ (predicted 71.92, 0.9σ against SH0ES in combined error). The derivation yields 7 unique predictions with explicit falsification conditions (Table 11). Since Paper I obtained the electromagnetic coupling constant $\alpha = 1/137.036\,082$ from the same signature (5, 2), the two forward chains on the same 4 axioms share one sign structure (5, 2) – that the electromagnetic coupling constant and the metric of gravitational quantization descend from the same root is the structural conclusion of this series.

Keywords: Wheeler-DeWitt equation; DeWitt metric; problem of time; superspace; metric signature SO(5,1); conformal mode; quantum cosmology; Compare-And-Swap; computational primitives in physics

1. Introduction

The fundamental equation of canonical quantum gravity is the Wheeler-DeWitt equation

$$\hat{H} \Psi[h_{ij}] = 0 \quad (1)$$

[1,2]. Here Ψ is the wave function of the universe, a functional of the spatial 3-metric h_{ij} , and $\hat{H}$ is the Hamiltonian constraint. The equation carries two standard structures. First, the coordinate time t does not appear in it – this staticity, under which the state of the entire universe appears “frozen,” has remained unresolved for 59 years since 1967 under the name of the “problem of time” [3,4]. Second, the DeWitt metric that defines the kinetic term,

$$G_{ijkl} = \frac{1}{2} h^{-1/2} (h_{ik} h_{jl} + h_{il} h_{jk} - h_{ij} h_{kl}) \quad (2)$$

is defined on the 6 independent components of the symmetric 3-metric and has pointwise signature $(-, +, +, +, +, +) = (5, 1)$ [1]. Why the components number 6 and why the signature is $(5, 1)$ rather than anything else – in particular, why there is exactly one negative direction and why it is exactly the scale (conformal) mode – has received no axiom-level account since 1967, and has persisted in Euclidean quantum gravity as the open problem known as the conformal-mode instability [26].

These structures have been organized into three criticisms:

- **(R1) Absence of time:** Why does t disappear from the equation, and what is the relation between the vanished time and the time we experience (the problem-of-time taxonomies of Kuchar [3] and Isham [4])?
- **(R2) Missing origin of the signature:** Why is the DeWitt metric $(5, 1)$, and why is the negative axis (the conformal mode) unstable [1]?
- **(R3) Measure and boundary conditions:** The Klein-Gordon-type inner product of Ψ yields negative probabilities, normalization generally fails, and the boundary condition is contested between the no-boundary proposal [7] and the tunneling proposal [8] – the equation itself selects neither.

These criticisms have remained unresolved for more than half a century.

The present paper answers the three criticisms – R1 and R2 by derivations inside the axiomatic system, R3 by a structural diagnosis binding the three symptoms to one root (with the two key equivalence proofs explicitly marked incomplete). We show that the same 4 axioms as Paper I [5] force the four constituents of Eq. (1) uniquely to the domain bit reading $(0, 1, 0, 1)$ (Theorem 1), that the absence of time is the operational mechanism of level-set projection (Theorem 2, R1), and we derive the component count 6 and the signature $(5, 1)$ of the DeWitt metric from the axiomatic structure (Section 4, R2). The three symptoms of R3 (negative probability, absence of measure, boundary-condition contest) are shown item by item to be symptoms of one and the same root – the axes switched off in the reading (Section 6).

Position of the present paper. The present paper is the second in the Axiomatic Model series – Paper I [5] (hereafter Paper I) treats the electromagnetic constant, the present paper treats the metric of gravitational quantization, following the same convention by which the source we cite as [1] was itself the first paper of a series, “Quantum Theory of Gravity. I.” The complete axiomatic system comprises 15 axioms [6]. The present paper uses *only 4 axioms and 1 proposition* of them to produce the above. Paper I [5] showed that within the same axiom set the irreversible/reversible classification of cost forces the quadratic-form signature $(5, 2)$ (Paper I, Theorem 1), and reached the fine-structure constant $\alpha = 1/137.036\,082$ (relative deviation 6×10^{-7} against experiment; the status of this residual and the α^3 -order correction task are stated in Paper I, Section 8.6) via the bounded symmetric domain D_5 .

The present paper treats the second consequence of the same (5,2): when, of the two times of the Banya equation, the domain time is cut by now-fixing – the same operation read in the cost basis as (5,2) – observer – the signature (5,1) of the DeWitt metric appears. As Paper I supplied an axiom-level ground for the half-century unresolved sign (5,2) of Wyler’s formula, the present paper supplies a ground of the same level for the sign (5,1) of the DeWitt metric – the two signs descend from one sign structure on the same 4 axioms.

Status classification. As in Paper I [5], the epistemic status of every result is classified into four levels: *I. Axiom-derived* (formal proof or forward chain completed), *II. Axiom-determined* (the axioms force the structure, which is then mathematically determined), *III. Axiom-described* (derivation path and interpretation are given in the text; independent formal proof is reserved for later papers), *IV. Out of scope* (not included in the scope of the present paper). Table 9 catalogues all results of the present paper under this classification.

Organization of the present paper: Section 2 presents the axiomatic system (the same set as Paper I, with the items actually used made explicit). Section 3 proves the three placement theorems (uniqueness of the bit reading, level-set projection, the $\delta = 0$ translation). Section 4 derives the component count and the signature of the DeWitt metric. Section 5 presents the structural consistency check – the ignition of the two switched-off axes overlaps one-to-one with the two workaround families of the literature. Section 6 answers the three criticisms item by item. Section 7 presents auxiliary results (loop-quantum-gravity numerical approaches, the Hubble-tension forward item). Section 8 discusses. Section 9 concludes.

The constraints imposed by the axioms are powerful. The four axioms of the present axiomatic system declare, respectively, orthogonality, atomicity, minimal cost, and causality preservation. These constraints force a logical minimal convergence: the structure satisfying the constraints converges to a unique point, and that point coincides with physical constants and with established mathematical structures. Because the deduction starts from constraints, there are no free parameters and fitting is impossible; because the search space is closed by the axioms, numerology is blocked. The constraints determine the output.

We show that axioms describe nature. The Banya equation was designed to be compatible with existing physics (see Paper I [5], Section 2, methodological transparency). That compatibility is a design criterion, not a discovery. The non-trivial claim is that the cost structure of this compatible equation produces, without fitting, specific forms matching experiment and the established numerical structures of existing theory. In Paper I that form was $\alpha = 1/137.036\,082$. In the present paper the forms are $\dim = 6$, the signature (5,1), and the identity of the negative axis.

2. Axiomatic System

2.1. Relation to Paper I

The axiom set used in the present paper is exactly identical to Paper I [5], Section 2: Axiom 1 (the Banya equation – 4-axis orthogonal norm), Axiom 2 (the conditional state transition CAS is the unique operator), Axiom 3 (cost – crossing an orthogonal boundary + in the prescribed order incurs cost +1), Axiom 4 (δ is the global flag), Proposition 1 (complete descriptive degrees of freedom $7 = 4 + 3$), Definition 1 (cost enumeration $13 = 8 + 5$), Definition 2 (ball value $4 = 1 + 3$, residual cost $9 = 13 - 4$). The full text of each item, the exhaustive 12-item list, the methodological-transparency argument (zero free parameters, 10 admissible numbers, closed search space), and the physical translation table of operational terminology follow Paper I, Section 2, and are not repeated here. The complete 15-axiom system is in [6].

2.2. The 4 Explicit Axioms + 1 Proposition (Summary Reproduction)

The correspondence between the axiom numbers of the present paper and those of the complete system [6] is the same as in Paper I:

Table 1. Axiom number correspondence.

Present paper	Complete system [6]	Content
Axiom 1	Axiom 1	Banya equation (4-axis orthogonal norm)
Axiom 2	Axiom 2	CAS as unique operator
Axiom 3	Axiom 4	Cost
Axiom 4	Axiom 15	δ global flag

When the text cites other axioms of the complete system (Axioms 6, 8, 10, 11, 12, 13, etc.), the notation “[6] Axiom n ” is used to distinguish them from the numbering of the present paper. Two notes. First, Axiom 4 of the present paper is an abridgment that reproduces only the δ -global-flag part of Axiom 15 of the complete system – the 128-state structure of δ in Axiom 15 is not included in the present paper, and this is the ground on which Section 8.6 places the formal derivation of the Born rule out of scope (IV). Second, the axiom labels of the abstract (orthogonality, atomicity, minimal cost, causality preservation) inherit the naming of Paper I [5] verbatim.

Axiom 1 (Banya equation – 4-axis orthogonal norm). *Every change δ of the universe is the norm of 4 mutually orthogonal axes:*

$$\delta^2 = (\text{time} + \text{space})^2 + (\text{observer} + \text{superposition})^2. \quad (3)$$

The first bracket is called the classical bracket (DATA), the second the quantum bracket (OPERATOR). The two brackets are orthogonal. The δ on the left-hand side is the global quantity of change that pervades the whole system (system time: 1 cycle = 1 tick), while the time axis on the right-hand side is the local domain axis inside the classical bracket – this left/right distinction is an immediate consequence of Axiom 1, and the names system/domain time together with the logarithmic relation are registered in the propositions of Axiom 2 and Axiom 15 of the complete system [6]. It is used throughout the present paper.

Axiom 2 (The conditional state transition (CAS) is the unique operator). *Every change of the universe is the repetition of a single conditional state transition. CAS proceeds in 3 steps – observe (Read), compare (Compare), record (Swap) – which are mutually orthogonal and whose order is forced (in the complete system [6] the enforcement of order is the role of the lock of Axiom 5; this summary follows Paper I [5] in incorporating the relevant content of Axiom 2 and Axiom 5 together). CAS operates in the OPERATOR bracket and records its result into the DATA bracket at the record step.*

Axiom 3 (Cost). *Crossing an orthogonal boundary + in the prescribed order incurs cost +1. In the absence of order the cost is 0. Cost is a dimensionless counting measure tallying ordered-boundary crossings.*

Axiom 4 (δ is the global flag). *The left-hand δ of Eq. (3) is outside the 7-bit internal machine that the right-hand side defines. If $\delta = 1$ the entire right-hand side is valid; if $\delta = 0$ it is void.*

Proposition 1 (Proposition 1 – complete descriptive degrees of freedom). *The total number of internal degrees of freedom of the system is $4 + 3 = 7$, as a direct consequence of Axiom 1 (4 axes) and Axiom 2 (3 steps). This 7 is also expressed by the triangular-number identity $7 = T(3) + 1$ ($T(3) = 6$ is the triangular number; +1 is self-reference). The identification of this +1 is derived in Lemma 2 of the present paper on the ground of [6] Axiom 10 – a different tier of bundling from the complete system’s $137 = T(16) + 1$, where the +1 is assigned to the self-reference of δ (the equals sign).*

Note: since this proposition is a direct logical consequence of Axiom 1 and Axiom 2, it is an auxiliary proposition rather than an independent axiom (identical to the “demoted to Lemma” annotation of Paper I [5], Table 3); textual references are standardized as “Proposition 1.”

2.3. Results of Paper I Used Here, and the Exhaustive Item List

Paper I, Theorem 1 [5]: under Axioms 1-4 the cost classification of the 7 axes is unique, and the partition into 5 irreversible axes (time, space, R, C, S; cost > 0) and 2 reversible axes (observer, superposition; cost $= 0$) forces the quadratic-form signature (5,2). Of the 8 possible partitions the remaining 7 are formally excluded. The present paper uses this theorem as input and does not re-prove it.

For methodological transparency we list exhaustively the items actually used in deriving the results of the present paper: Axioms 1-4 (reproduced in Section 2.2), Proposition 1 (complete descriptive degrees of freedom, Section 2.2), Definition 1 (cost enumeration $13 = 8 + 5$ – the source of the 8 of Section 7.1), Definition 2 (ball value $4 = 1 + 3$ – the source of the $1/4$ of Section 7.1 and of the multiplier correspondence of Section 8.3), Definition 3 (bit reading, newly introduced in the present paper, Section 2.4), Paper I Theorem 1 (the paragraph above; the input of Section 4.3), and two registered statements of the complete system [6] (the self-reference channel of Axiom 10 – the ground of Lemma 2; the appendix statement of the Axiom 15 precinct – the explicit premise of Theorem 2 and the source of the structural-view (5,2) and subframe-(3,1) statements of Section 4.3, Route 1). No independent assumptions beyond these are used, and there are no free parameters.

2.4. Definition of the Domain Bit Reading

Definition 3 (Bit reading). *For the 4 axes of Axiom 1 (observer, superposition, time, space), the bit reading of a physical equation is defined by the following correspondence rule: if a constituent of the equation explicitly requires an axis, the corresponding bit is 1; if no constituent of that axis appears in the equation, it is 0. The reading is written as the 4-bit tuple (observer, superposition, time, space). A constituent that appears notationally but can be eliminated without changing the content of the equation is not a requirement. The reading is defined with respect to the presented form of the equation; the present paper uses the standard canonical presented form of Eq. (1).*

This reading is not an interpretation but a constituent inspection – that each bit is determined independently by a different constituent of the equation is the content of Theorem 1. The uniqueness of Theorem 1 is uniqueness under this correspondence rule, and the rule itself is declared in advance as Definition 3.

2.5. Supplementary Terminology

The physical translation of operational terminology follows Paper I [5], Table 2; the terms newly introduced by the present paper are the following.

Table 2. New terms of the present paper.

Term	Definition	First use
bit reading	Binary inspection of whether the constituents of an equation explicitly require each of the 4 axes	Definition 3
level-set projection	The operation by which the composition of now-fixing and rearrangement leaves a single level set of the time function	Theorem 2
ignition	Switching a bit that is off in the reading back to 1, moving to a different formalism	Section 5
excision	Removal of the observer axis. The substance of the -1 in the component count $7 - 1$ and of the $-$ operation in the signature $(5, 2) - \text{observer}$	Sections 4.2, 4.3
cut	Fixing one time axis to now, turning it into a level set. The operation by which the signature loses one negative axis	Section 4.3, Route 1
observer-absent slice	The subspace of reading $(0, 1, 0, 1)$. The same configuration as 1010 in the string convention of the complete system	Section 3, notation note
unfired	The state in which $\delta = 0$, CAS has not been engaged, and there is no DATA record	Theorem 3
screen rendering	The recording of a determinate state into DATA upon a $\delta = 1$ firing (a term of Paper I [5])	Corollary 3.1

2.6. Reader's Guide: Four Common Misconceptions

We state in advance the points of the present paper's notation and claims that are prone to misreading:

1. *The absence of time is not the non-existence of time:* by Theorem 2 the elimination of time is a level-set projection, and the axis that assigns the level-set values persists outside the equation as system time (the left-hand side of Axiom 1) (Corollary 2.1).
2. *The right-hand-side zero is not an energy cancellation:* under substitution of real values the Banya norm is a sum of two squared terms, so no opposite-sign cancellation path exists, and the zero translates only as $\delta = 0$ (unfired) (Theorem 3).
3. *The $-$ in $(5, 2) - \text{observer}$ is not arithmetic subtraction:* the negative sign of the cost classification is the type marker of the filter dimensions (Paper I [5], Theorem 1 (B)), and the $-$ observer of the present paper is the excision of one reversible axis, a set operation (Section 4.3).
4. *The conformal-mode instability is not a defect of the theory:* that the remaining negative axis stays timelike is the scar of the observer-excision surgery, a cross-section of the balance that the full signature $(5, 2)$ maintained (Section 4.4).

3. Placement Theorems – The Axiomatic Coordinates of the Wheeler-DeWitt Equation

This section fixes the position of Eq. (1) within the axiomatic system by three theorems. This position is the input of the derivation in Section 4.

Theorem 1 (Uniqueness of the bit reading). *Under the standard canonical presented form of Eq. (1) and the correspondence rule of Definition 3, the constituent configuration of the Wheeler-DeWitt equation is uniquely the domain bit reading (observer, superposition, time, space) = $(0, 1, 0, 1)$.*

Proof. Apply the correspondence rule of Definition 3 to the constituents of Eq. (1). Four verdicts force the four bits independently.

(i) *superposition = 1:* $\Psi[h_{ij}]$ is an object in which all configurations of the 3-metric coexist simultaneously as undetermined. In Axiom 1 the only seat of undetermined multiplicity is the superposition axis. With superposition = 0 the object called a wave function is not even defined.

(ii) *space = 1:* the argument h_{ij} is the spatial 3-metric, a configuration on the recording surface. In Axiom 1 the seat of recording is the space axis of the classical bracket. With space = 0, Ψ has no domain of definition. Here space = 1 means the opening of the recording surface (the existence of

the domain of definition), not the occurrence of recording – whether recording occurs is a matter of δ , treated in Theorem 3.

(iii) $time = 0$: the coordinate time t does not appear in the equation. By Definition 3, a formalism with $time = 1$ is any formalism in which t appears as an explicit variable – time-dependent equations of the form $i\hbar \partial_t \Psi = \hat{H} \Psi$ (Schrödinger, Tomonaga-Schwinger) being the representative cases – and Eq. (1) differs from all of them at the constituent level. The mechanism of the elimination is supplied by Theorem 2.

(iv) $observer = 0$: no observer appears in the equation. By Definition 3, a formalism with $observer = 1$ is any formalism in which an observer appears as an explicit constituent – relative-state formalisms that condition on an observer (Page-Wootters conditional probability [9]) being the representative case – and Eq. (1) differs from all of them at the constituent level.

Since the four verdicts arise independently from different constituents (the existence of Ψ , the argument h_{ij} , the absence of t , the absence of an observer), the configuration is uniquely $(0, 1, 0, 1)$ under the correspondence rule of Definition 3. The exhaustive exclusion of the remaining 15 combinations is in Table A1 of Appendix A; representative exclusions are summarized in Table 3. The correspondence structure of the four verdicts is shown in Figure 1. $\square$

Table 3. Representative exclusions of domain bit combinations – the exhaustive 16 are in Appendix A. Notation: (observer, superposition, time, space).

Combination	Corresponding formalism	Verdict
(0, 1, 0, 1)	Wheeler-DeWitt $\hat{H}\Psi = 0$	Adopted – all four verdicts satisfied; sole survivor
(0, 1, 1, 1)	Time-dependent Schrödinger / Tomonaga-Schwinger	Excluded – violates verdict (iii)
(1, 1, 0, 1)	Page-Wootters conditional-state formalism [9]	Excluded – violates verdict (iv). The ignition direction of Section 5 (i)
(1, 1, 1, 1)	Effective description of the fully fired state (screen physics)	Excluded – violates verdicts (iii), (iv) simultaneously
(0, 0, 1, 1)	Classical spacetime (Einstein equations)	Excluded – violates verdicts (i), (iii); no Ψ
(0, 1, 0, 0)	Pure superposition with no recording surface	Excluded – violates verdict (ii); no domain of definition

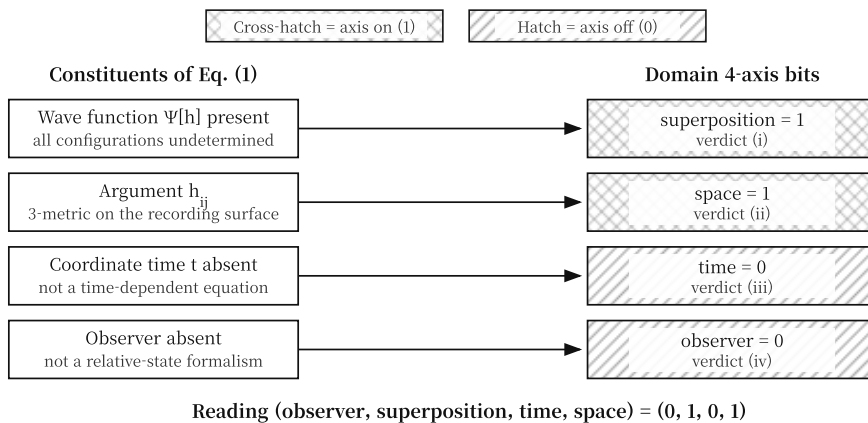


Figure 1. The bit reading of Theorem 1 – four constituents force four bits independently. Cross-hatch marks an axis that is on (1), hatch an axis that is off (0).

Note on notation: the bit assignment of the complete system [6] is bit 0 = observer, bit 1 = superposition, bit 2 = time, bit 3 = space, and its string convention writes from the highest bit (bit 3 =

space) – in that convention the present configuration is written 1010. The present paper standardizes instead on the axis-ordered tuple of Axiom 1, (observer, superposition, time, space) = (0, 1, 0, 1). The two notations denote the same configuration; hereafter the text calls it the “observer-absent slice.”

Theorem 2 (Time elimination = level-set projection). *The composition of now-fixing and rearrangement leaves a single level set of the time function, and on the level set the axis that assigns the level-set values drops out by projection. Hence the absence of time in $\hat{H}\Psi = 0$ is not the non-existence of physical time but a structural consequence of the operation.*

Premise (reading): the formulation of the Hamiltonian constraint – a construction that leaves no explicit time variable – is read in the present system as the composition “now-fixing + rearrangement.” This reading is not an invention of this theorem but the appendix statement of the Axiom 15 precinct of the complete system [6] (the negative sign is now; fixing the present turns the two time axes into level sets), and is declared here as an explicit premise. In the standard understanding the absence of time is a consequence of the Hamiltonian constraint ($H \approx 0$) enforced by reparametrization invariance; the present reading re-reads that constraint structure as the operations of now-fixing and rearrangement – a restatement, not a replacement, of the standard origin. Of the two time axes, the one to which Eq. (1) applies is the domain time axis of the classical bracket, and it is that axis which Theorem 2 eliminates.

Proof. Under the above reading, three steps, each of which is standard mathematics. (1) *Now-fixing:* fixing the coordinate time t at its present value demotes t from an independent variable to a constant. (2) *Rearrangement:* moving all terms of the Hamiltonian constraint to one side yields the implicit form $F[h_{ij}] = 0$. (3) *Projection:* the implicit form of a variable fixed as a constant is a single level set of the family of functions whose values that variable assigns, and on a level set that variable’s axis is by definition invisible (Figure 2). $\square$

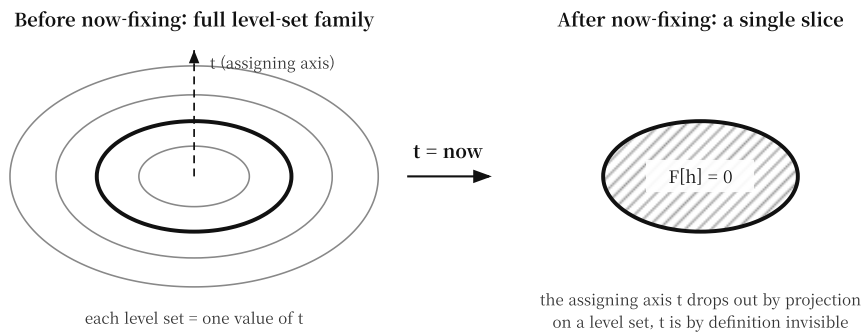


Figure 2. The level-set projection of Theorem 2 – the composition of now-fixing and rearrangement leaves a single level set of the time function (hatched), and the axis that assigns its values drops out by projection.

Corollary 2.1. Not a universe from which time has vanished, but an operation that has kept a single slice. That altitude cannot be asked on a contour line is not because the mountain has no altitude but because one contour line is by definition a single altitude. The mystery of the “frozen formalism” [3,4] dissolves under the present reading – not a mystery but the price of an implicit-function representation.

Theorem 3 (Uniqueness of the $\delta = 0$ translation). *Since the Banya norm is the positive-definite norm of the sum of two bracket squares, the right-hand-side zero of $\hat{H}\Psi = 0$ cannot be translated as a cancellation between kinetic and potential terms and translates only as $\delta = 0$ (unfired).*

Proof. The norm of Axiom 1, $\delta^2 = (\text{DATA})^2 + (\text{OPERATOR})^2$, is a sum of two squared terms at the two-bracket tier. Under substitution of real values each bracket square is non-negative, so for the sum to vanish each term must vanish, and no opposite-sign cancellation path exists. In the axiomatic system the state in which both terms vanish is the state in which $\delta = 0$ and CAS has not been engaged – the state in which only polling runs ([6] Axiom 8: the firing of δ itself costs 0), with the equality not holding = superposition maintained ([6] Axiom 10 proposition) and no DATA record: unfired. $\square$

Note on tiers: the positive-definiteness of this theorem is a statement at the *two-bracket* squared-sum tier of Eq. (3). The signature (5, 2) is a property of a different tier – the quadratic form of the sign function that the cost classification assigns to each of the 7 axes (Paper I [5], Definitions 4, 5) – and its negative (–) sign is not arithmetic subtraction but the type marker of the filter dimensions (see Paper I [5], Theorem 1 (B)). The two forms are different granularities of the same object, so the two-bracket positive-definiteness and the 7-axis (5, 2) do not contradict. Moreover, the Banya equation itself is a topological declaration, not a metric; metric notions such as sign and distance are determined only upon substituting constants ([6] Axiom 1 proposition) – the positive-definiteness of this theorem is a statement on top of exactly that metrization step of substituting real values.

Corollary 3.1. The staticity of the Wheeler-DeWitt equation is not a defect but the exact description of the pre-firing state. What the equation describes is not “a universe unchanging in time” but “the interval in which screen rendering has not yet begun.” This is the input of Section 6 (R3) and of Section 8 (the position of the no-boundary proposal).

4. Main Result – Derivation of the Component Count and Signature of the DeWitt Metric

4.1. The Object

At each point of superspace [2], the DeWitt metric (2) is defined on the 6 independent components of the symmetric 3-metric h_{ij} and has pointwise signature (5, 1) [1]. The mathematical derivation of the component count 6 and the signature (5, 1) was itself completed by the diagonalization of DeWitt 1967 [1] and generalized to the full λ -family by Giulini [25]. The question of the present paper is upstream of that – the axiom-level ground of why that structure – and what has not been supplied for 59 years is the ground at this tier. The derivation of this section is therefore a postdictive reconstruction of known facts (retrodictive derivation) and carries no predictive risk by itself – predictive risk is carried by the forward items of Table 11. This section derives the two numbers on top of the placement fixed by Theorem 1.

4.2. Component Count: $6 = T(3) = 7 - 1$

Lemma 2 (The channel of the self-referential +1 is observer). *In the identity $7 = T(3) + 1$ of Proposition 1, the only axis of the 7 that can be counted as the +1 is observer.*

Proof. Self-reference is the loop by which the global δ accesses its own output. Axiom 10 of the complete system [6] specifies the path of this loop uniquely: $\delta \Rightarrow \text{observer} \Rightarrow \text{CAS} \Rightarrow \text{result} \Rightarrow \delta$ – the only channel through which δ accesses itself is observer. Hence among the 7 internal axes the internal landing point of the self-referential loop is observer alone, and the only axis that can stand in the +1 position of the $T(3) + 1$ decomposition is observer. $\square$

Derivation. By Proposition 1 the total internal degrees of freedom are $7 = T(3) + 1$, and by Lemma 2 the seat of the +1 is observer. By Theorem 1 the slice on which the DeWitt metric lives is observer-absent, so what is excised is exactly that +1. What remains is

$$\dim G_{ijkl} = 7 - 1 = T(3) = 6. \quad (4)$$

Here 6 is not an axiom-registered number – it is the result of a structural operation ($7 - 1$) on the registered numbers 7 and 1, and this point is consistent with the filter of Paper I [5], Section 8 (6 itself factors as 3×2 and is barred from registration): precisely because 6 cannot be registered, it must be reached only by operations on registered numbers, and the present derivation supplies that path.

Comparison of explanatory scope. Three computations reach the component count 6 (Table 4). Since 6 is the number defined by the independent-component count of a symmetric 3×3 matrix (3 diagonal + 3 off-diagonal), $3 + 3$ is not a competing hypothesis but the explanandum (the definition) that any derivation must reproduce. The decisive question is which computation also explains the signature.

Table 4. Three computations of the component count 6 and their explanatory scope.

Computation	Content	Status	Explanatory scope
$3 + 3$	3 diagonal + 3 off-diagonal	Definition	The mathematical definition of 6 itself – the target every derivation must reproduce; carries no information about the sign structure
$\binom{4}{2} = 6$	Pairs of the 4 domain axes	Candidate – insufficient	Reproduces the count but is silent on the signature – no explanatory path for why exactly one axis is negative
$T(3) = 7 - 1$	Self-reference excised from the complete descriptive degrees of freedom	Adopted	Explains count and signature by one and the same excision operation. $7 = T(3) + 1$ is the registered identity of Proposition 1; $+1 = \text{observer}$ is Lemma 2; the excision is Theorem 1

4.3. Signature: The Double Derivation of (5, 1)

The hierarchy of three viewpoints. The present axiomatic system explains the signature from three viewpoints. The parent (5, 2) is explained from the axioms along two routes – the cost viewpoint (irreversible/reversible classification – Paper I [5] Theorem 1) and the structural viewpoint (geometric count of 5 including δ + two times – the appendix of the Axiom 15 precinct of the complete system [6]) – and since the two routes view the same norm in different orthogonal bases, both are forced to the same (5, 2) (the sign is an invariant of the norm [6]). For the step down from the parent to (5, 1), the time viewpoint – the now-cut of the domain time, one of the two times – is the most intuitive, so this section presents it as the main derivation (Route 1) and appends the observer excision of the cost viewpoint as a cross-derivation (Route 2). That several deductions from different directions reach the same conclusion is a strength of the Banya-equation structure: the two routes share the parent (5, 2) and Theorem 1, so they are not fully independent; theirs is the relation of a cross-validation drawing discriminating power from different bits of Theorem 1 – a robustness argument of the same shape as in Paper I [5], where α and $\sin^2 \theta_W$ emerged from the same axiom set along different forward chains.

Derivation Route 1 (time route – the cut of the domain time). The appendix of the Axiom 15 precinct of the complete system [6] registers the structural viewpoint of (5, 2) as follows: the positive (+) 5 axes are the geometric count in which the equals sign binds the global δ (1) with the 4 local axes (time, space, observer, superposition), and the negative (–) 2 axes are time – system time (1 CAS tick) and domain time (the rendered logarithmic time). The negative sign is now: fixing the present turns the two time axes into level sets, and on the level sets the 5 domains unfold hyperbolically. The independence of the two times is also registered at the source – the now of the left-hand side (system) and the now of the right-hand side (domain) are coupled by the equals sign without being forced equal, so the two times are two timelike directions on the level set.

On this geometry the theorems of the present paper become the knife. Since there are two time axes, there are two places where a now-fixing cut can be made, and which one is cut determines which famous signature results (Figure 3):

- *Cutting the domain time*: the t -elimination of the Wheeler-DeWitt equation (time = 0 of Theorem 1, mechanism the now-fixing of Theorem 2) is exactly this cut. When one of the 2 timelike directions is cut into a level set, what remains is 5 spacelike directions + 1 timelike direction (the system-time side): $(5, 1)$ – the DeWitt metric. The same arithmetic as the component count $7 - 1 = 6$ holds in the signature count.
- *Cutting the system time*: when the system time becomes a constant as the instantaneous value of polling, what remains on the screen (the classical bracket) is 3 positive space axes and 1 negative domain-time axis – $(3, 1)$, observable Minkowski spacetime. The source appendix registers this explicitly: “the subframe $(3, 1)$ is observable spacetime.” Its time value is the instantaneous value stamped as a timestamp (a record of the moment); numerically it is the present value of $t_{\text{dom}} = \log(T_{\text{sys}})$.

Table 5. Two cuts, two signatures – there are two times, hence two places to cut.

Time cut	Remaining metric	Signature	Remaining negative axis	Corresponding physics
System time (polling instant)	Screen metric – space of the instant	$(3, 1)$	Domain time	Minkowski spacetime (observable; the subframe statement of [6])
Domain time (the t -elimination of WDW, Theorem 2)	Superspace metric – configuration space of the instant	$(5, 1)$	Trace of system time	DeWitt metric (main result of the present paper)
No cut	Full extension	$(5, 2)$	Both times	The Wyler sign of Paper I [5] – the source of α

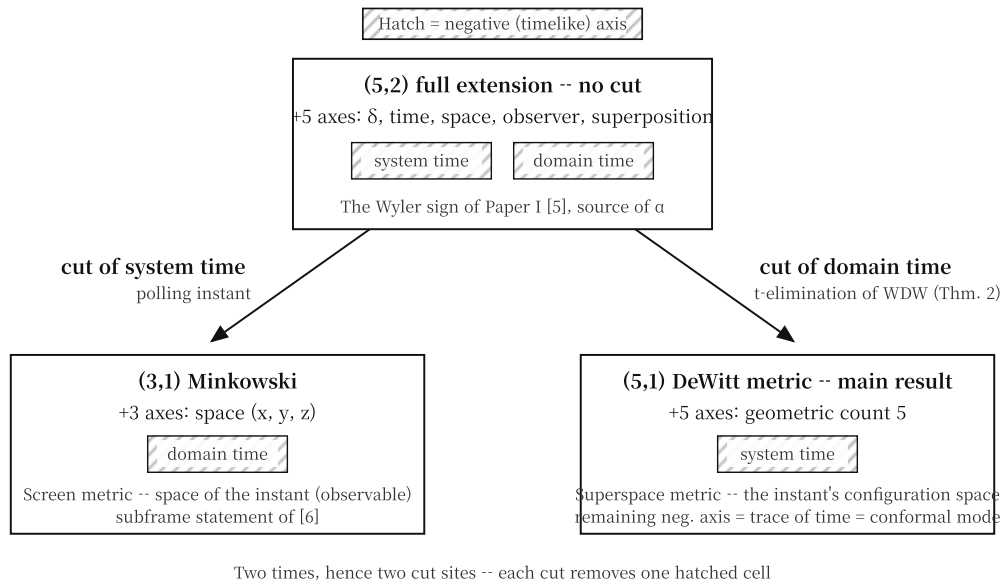


Figure 3. Two cuts, two signatures – the 2 negative axes of the parent $(5, 2)$ are the two times. Cutting the system time leaves observable spacetime $(3, 1)$; cutting the domain time (Theorem 2) leaves the DeWitt metric $(5, 1)$. The two famous Lorentzian signatures are the two complementary cuts of “which time is fixed to now.”

Both cuts are snapshot photography – they differ only in which time is frozen for the shot. Freezing the system time photographs the space of the instant (Minkowski); freezing the domain time photographs the configuration space of the instant (DeWitt superspace); the full extension with

nothing frozen is the (5, 2) of Paper I. That the two famous Lorentzian signatures of physics – the (3, 1) of observable spacetime and the (5, 1) of superspace – are the two complementary cuts of “which of the two times is fixed to now” is the point of this picture. Theorem 2 (level-set projection) is in this picture both the mechanism of time elimination and half of the signature determination – one act of now-fixing is the cut of one negative axis.

Derivation Route 2 (cost route – cross-derivation). By Paper I Theorem 1 [5], the cost classification of the 7 axes is unique, 5 irreversible (+) / 2 reversible (–), with signature (5, 2) – Paper I explicitly excluded the alternative 21-dimensional groups SO(7) and SO(4, 3) in this cost-category derivation, and the cost classification is determined axis by axis by the axioms. By Theorem 1 of the present paper, the slice on which the DeWitt metric lives is the subspace in which observer, one of the 2 reversible axes, is absent. The signature with one reversible axis excised is

$$(5, 2) - \text{observer} = (5, 1). \quad (5)$$

The 5 positive axes coincide in full with the 5 irreversible axes of Paper I Theorem 1 (time, space, R, C, S).

The physical meaning of Route 2 – the observer’s self-exclusion. The –observer of Eq. (5) is not peculiar to the Wheeler-DeWitt equation but the universal structure of the act of observation. What is observed is the output that has passed the observer filter, and a filter cannot place itself on its own list of things passed – the same self-referential structure as an eye that cannot see itself. The axiomatic grounds are already registered: δ cannot reach DATA directly and must pass the observer filter ([6] Axiom 13); observer is the subject of access, not its object ([6] Axiom 10); and the reversible axes “only select which data to fetch or which branch to take – they are never themselves recorded as data” (Paper I [5], Theorem 1 (B)). Hence the signature of the observable world is always the observer-less (5, 1), and its screen cross-section is (3, 1) – while (5, 2) is the whole that includes the observer itself, an extension never captured in its entirety by any observation.

That the two negative axes of the same bracket are eliminated in different ways is the crux of this structure. The observer is excluded *in principle* – being the subject of observation, it is forever absent from the observation list. Superposition is eliminated *in execution* – when an observation is actually carried out (Compare true), collapse occurs and the superposition is recorded into DATA as a single determinate state ([6] Axiom 7). The former is an ever-holding exclusion, so the standing signature of the observable world is (5, 1); the latter occurs per observation event, so in a completed record even the remaining negative axis disappears – the consequence in that direction is treated as a hypothesis in Section 8.2.

Compatibility of the two routes and the distinction of the two counts: the basis of the time route (Route 1) and the basis of the cost route (Route 2) view the same norm in different orthogonal bases, and since the sign is an invariant of the norm they reach the same (5, 1). Moreover, the 5 of Route 1 is the geometric count including the global δ , while the 7 of the component-count derivation (Section 4.2) is the local degrees-of-freedom count – the source states explicitly that “what separates the two is the equals sign, and the two numbers do not conflict.”

Note on where the discriminating power resides: counting alone, excising any of the 7 axes leaves 6, and excising either of the 2 reversible axes yields (5, 1). The discriminating power of Route 1 – that the time being cut is none other than the domain time – therefore comes from the bit reading of Theorem 1 (time = 0); and the discriminating power of Route 2 – that the excised axis is none other than observer – comes from the same Theorem 1’s bit reading (observer = 0, superposition = 1) together with Lemma 2 (+1 = observer). The two routes draw their discriminating power from different bits of the same Theorem 1 and arrive at the same (5, 1), so this convergence is itself a cross-validation of the bit reading. Section 4.4 (the identity of the remaining negative axis) is a further test: were the excised axis wrong, the identity of the remaining negative axis would change and the correspondence with the conformal mode (Table 6) would break.

4.4. The Identity of the Negative Axis – Reinterpreting the Conformal-Mode Instability

That the single negative direction of the DeWitt metric is exactly the scale (conformal) mode is a standard fact [1], persisting in Euclidean quantum gravity as the open problem of the conformal-mode instability. In the present derivation the remaining reversible axis is superposition alone – this assignment itself is a derivation (formal consequence) of Section 4.3. On top of it we propose the following correspondence: superposition is, in the axiomatic system, addressless indexing – the axis that unfolds all possible states at once ([6] Axiom 13, the RLU addressless angular index) – and the character of this axis, pointing at the whole set of states at once, coincides with the character of the conformal mode, which changes the overall scale of the metric at once. Deriving the identity of indexing and scale formally from the axioms is future work; the status of this correspondence is not a derivation but an interpretive proposal on top of the (formal) axis assignment – the corresponding card of the complete system [6] records it at the same level (an observed correspondence).

Table 6. Axis correspondence, summary.

DeWitt-metric side	Axiom side	Status
5 positive directions	5 irreversible axes (time, space, R, C, S)	Full coincidence with Paper I Theorem 1
1 negative direction = conformal mode	Remaining reversible axis = superposition	Axis assignment is the derivation of Section 4.3; the conformal correspondence is the interpretive proposal of Section 4.4
Absent	observer (the self-referential +1, excised)	Theorem 1 bit reading. The -1 of the component count $7 - 1$

The time route (Section 4.3, Route 1) reads the same negative axis in the other basis and yields a stronger consequence: the negative axis that was not cut is the trace of system time, a residual time axis. Then what quantum cosmology has adopted as convention – using the scale factor a as internal time in minisuperspace – becomes not convention but necessity: the negative (scale) direction of the DeWitt metric was originally time, and that is why it plays the role of time. Where the standard formalism has recorded that “the timelike direction happens to coincide with the scale mode,” the time route reads it the other way: “the remaining time is rendered into the seat of scale.” The cost-basis name (superposition, the scale index) and the structural-basis name (residual time axis) are two names of the same direction, and this coincidence follows from the compatibility of the two bases (the sign is an invariant of the norm).

Hence the conformal-mode instability is not a defect of the theory but the scar of the observer-excision surgery. That one reversible axis remains timelike in the subspace from which the observer has been excised is a cross-section, revealed by one excision, of the balance that the full signature (5, 2) maintained.

4.5. The Derivation Chain, Assembled

Axioms 1-4 + Proposition 1 $\Rightarrow$ (Theorem 1) $\Rightarrow$ $(0, 1, 0, 1) \Rightarrow$ (Theorems 2, 3) $\Rightarrow$ $\delta = 0$ level set $\Rightarrow$ (Section 4) $\Rightarrow$ $\dim = 6$, double derivation of (5, 1) (cost: (5, 2) – observer / structure: cut of domain time)

Connection to Paper I: the chain of Paper I [5] was $(5, 2) \Rightarrow \text{SO}_0(5, 2) \Rightarrow D_5 \Rightarrow \alpha$. The chain of the present paper is $(5, 2) \Rightarrow$ observer excision $\Rightarrow (5, 1) = \text{DeWitt metric}$. The two chains share the same root, Paper I Theorem 1 (Figure 4).

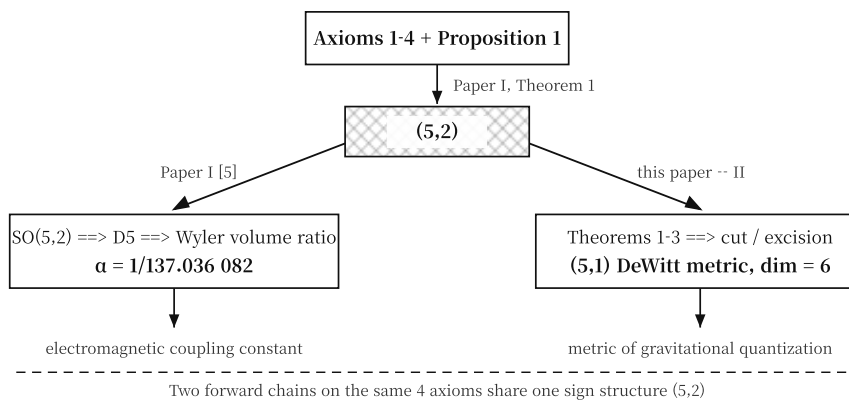


Figure 4. The two forward chains of the series – one (5,2), forced by the same 4 axioms, descends in Paper I to α and in the present paper to the (5,1) of the DeWitt metric.

5. Structural Consistency Check: Igniting the Two Switched-Off Axes

We present a second, independent structural check within the same axiomatic system. In Paper I [5] this role was played by $\sin^2 \theta_W$.

In the reading (0, 1, 0, 1) of Theorem 1 the switched-off axes are observer and time, two of them. If the reading is correct, a strong structural prediction follows: any attempt to recover time in Eq. (1) must take the form of switching exactly one of these two axes back on, and hence the workarounds must fall into exactly two families. The standard workarounds of the literature do exactly that (Figure 5):

- **(i) Observer ignition – the Page-Wootters family [9].** Designating part of the system as a clock and conditioning the rest on it recovers dynamics inside conditional probabilities. In the language of the axiomatic system this is switching on the observer axis and moving to (1, 1, 0, 1) – the $N = 2$ minimal case of multiple projection in the complete system ([6] Axiom 11). This is precisely the slot excluded in Table 3 as the violation of the observer verdict (iv).
- **(ii) Time ignition – the thermodynamic-arrow family [10,11].** The family that recovers the direction of time from entropy increase. In the language of the axiomatic system this is reading the direction of the domain time axis off the accumulation of cost (the geometric decay of RLU eviction, [6] Axioms 6, 12) – switching on the time axis and descending toward the screen-side description.

We reinforce the details of the two families with the records of the complete system [6]. In (i), the reading after observer ignition is (1, 1, 0, 1) and the time bit is still 0 – that is, conditioning on a clock subsystem is not a recovery of the time axis but an operation that picks one of the level sets that Theorem 2 pressed down by projection and locally re-stacks the domain. Clock and system are not separate entities but one global δ projected through two observers (the $N = 2$ minimal case, [6] Axiom 11), and the procedure of polling one projection (the clock) to read the other is precisely the recovered time. The experiment of Moreva et al. [24], demonstrating the internal/external-observer double mode of an entangled photon clock, reads as a laboratory illustration of this two-tier structure (a demonstration compatible with several formalisms, not evidence specific to the present system). In (ii), the low-entropy initial condition that the literature inserts as a separate postulate comes free in the present system: at the first tick of system time ($T_{\text{sys}} = 1$) the cache cannot but be empty, so the initial low entropy is not an assumption but a definitional consequence. This point is the root of Prediction 1 of Table 11 (no reversal upon recollapse) and the single point at which the prediction diverges from the Kiefer-Zeh family [10], which treats the arrow as a function of the scale factor.

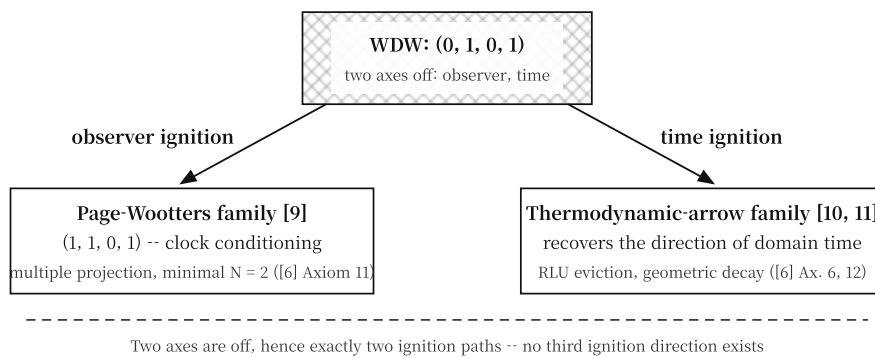


Figure 5. Igniting the two switched-off axes – the two time-recovery workaround families of the literature overlap one-to-one with the two axes switched off by Theorem 1. That there are only 2 ignition directions is a formal consequence of the reading; the placement of the two families is an interpretive mapping.

Completeness as an axiom consequence extends exactly this far: two axes are off, so there are exactly two ignitable paths (formal consequence). That the other approaches of the literature are placed into these two families is an interpretive mapping – the Brown-Kuchar dust clock [27], unimodular time [28], and York internal time are internal-time-school moves that designate some variable as time (first row of Table 8), an axis re-selection inside the slice rather than an ignition, and WKB emergent time is absorbed as the logarithmic-compression consequence of Section 8.3. Under this placement the two workarounds overlap one-to-one with the two switched-off axes, and this accords with the terrain that the problem-of-time taxonomies of Kuchar [3] and Isham [4] assembled empirically.

6. Answers to the Three Criticisms

6.1. (R1) Absence of Time: “Where Did Time Go”

Answer: Theorem 2 supplies the mechanism. Now-fixing + rearrangement = level-set projection. Not a universe without time but a single slice. What is lost is not time but the axis that assigns the level-set values, and that axis persists outside the equation as system time (the firing counter of δ , the left-hand side of Axiom 1). As Section 5 showed, the existence and the classification of the two families of time-recovery workarounds are the independent check of this answer.

6.2. (R2) The Signature: “Why (5, 1)”

Answer: by Section 4, the component count $6 = T(3) = 7 - 1$ and the signature (5, 1) are forced from the axioms – and by two routes drawing their discriminating power from different bits of Theorem 1 (time: the now-cut of the domain time / cost: (5, 2) – observer), converging on the same value. Once the sign is determined, the remaining negative axis follows as superposition, and the correspondence with the conformal mode (the interpretive proposal of Section 4.4) becomes the test of that identification. The conformal-mode instability is not a defect but the scar of observer excision – just as Wyler’s (5, 2) was no arbitrary choice in Paper I [5], DeWitt’s (5, 1) is not arbitrary either.

6.3. (R3) Measure and Boundary Conditions: Three Symptoms, One Root

All three symptoms of R3 are symptoms of the axes switched off in the reading (0, 1, 0, 1).

Negative probability. The negative probability of the Klein-Gordon-type inner product is not an error but an error message: probability is the screen projection of δ ’s free selection (the Born rule – formally derived in the complete system [6]) and requires an observer, but the observer = 0 slice has no screen to receive the projection. That the negative contribution to the inner product comes exactly from

the negative axis of the DeWitt metric – the conformal direction whose identity Section 4.4 pinpointed – is a standard fact consistent with this diagnosis. The details of the diagnosis are as follows [6]: in the present system the seat of probability is not the norm of a wave function but the tally of δ selections, and tallying requires a tallying subject (observer, the recipient of the Compare branch). Setting the superspace norm in that seat on the observer = 0 slice is probabilizing DATA (h_{ij}) directly without passing through observer – a severing of the self-referential loop ([6] Axiom 10) – and the negative probability is the symptom of that severing. The formalization that the Klein-Gordon-type inner product is indefinite and its probability interpretation fails is in Kuchar [3]. That switching observer on (Page-Wootters conditioning) restores non-negative conditional probabilities is likewise consistent with the same diagnosis. However, the equation-level equivalence proof between Page-Wootters conditional probability and the tally frequencies of the present system is recorded as incomplete in [6].

Non-normalizability. $\int |\Psi|^2 \mathcal{D}h = \infty$ is not a failure of the measure but the absence of one – in the $\delta = 0$ interval (Theorem 3) there is no screen rendering, hence no event space on which a probability measure could be defined at all. Normalizability is a property demanded only after firing ($\delta = 1$, the conditioned description).

Boundary-condition contest. The contest between the no-boundary proposal [7] and the tunneling proposal [8] is an extra input the equation does not determine. In the present system this is natural: the boundary condition is the junction information connecting the $\delta = 0$ description to the $\delta = 1$ world, so it cannot but lie outside the equation (the internal description of $\delta = 0$). The observation that the sign difference of the two proposals corresponds to the two reading directions, cost/norm (Paper I [5], Section 2, Common Misconception #3), is recorded at hypothesis level in [6]. DeWitt's own boundary condition $\Psi^{[(3)\mathcal{G}]} = 0$ (vanishing on singular 3-geometries) [1] is consistent with the discrete ring floor of the present system – the distance ℓ between two local states has minimum $\ell = 1$, $\ell = 0$ being unreachable in a discrete system ([6] Axiom 11 proposition), and singularities do not exist ([6] Axiom 13 proposition, grounded in the DATA discreteness of Axiom 3) – and the vanishing of the wave function on an unreachable configuration is not a condition but a definition. The λ -generalization of the DeWitt metric (anisotropic-scaling gravity) is in Horava [13], and it is the point of contact for the still-incomplete derivation of why $\lambda = 1$ is forced by the axioms (Section 8.6).

7. Auxiliary Results

7.1. Independent-Domain Numerical Approaches: Loop Quantum Gravity

The derivations of the present paper (Sections 3, 4) are structural and produce no continuous numbers. As an independent-domain comparison of the same status as the Hamming [7,4,3] coincidence of Paper I [5], Section 7.1, we present one numerical approach to the number of an unrelated school (the Barbero-Immirzi γ) together with its downstream item and one constituent-count reinterpretation (the area gap, QNM). Loop quantum gravity (LQG) [17] is the other line of canonical quantum gravity – holonomy variables instead of the superspace of the DeWitt metric – and has no causal connection whatsoever with the present axiomatic system.

Table 7. Loop-quantum-gravity numbers and the axiom-number approaches – derivation details in the corresponding cards of [6].

Quantity	LQG standard	Axiom-number expression	Source of the constituent numbers
Barbero-Immirzi parameter [14,15]	γ – a free parameter the theory does not determine. Root of the transcendental equation matching black-hole entropy $S = A/4$: $\gamma = 0.237533$ [20, 21]	$\gamma = \frac{\sqrt{3} \ln 3}{8} = 0.237857$ (0.136% against the literature value)	$\sqrt{3}$: workbench norm of the 3-axis CAS orthogonality (Axiom 2 proposition), $\ln 3$: state information of the 3 CAS steps, 8: the 8 bits of the d-ring and the read cost (Definition 1)
Area gap	$\Delta A = 4\pi\gamma\sqrt{3}l_p^2$	$\Delta A = \frac{3\pi}{2} \ln 3 l_p^2$	Downstream of γ above – a substitution identity
Black-hole quasinormal modes (QNM) [16]	Real part $\ln 3/(8\pi)$ (Hod), imaginary spacing $1/4$	$\ln 3$: 3 states, 8π : read $8 \times$ hemispheric arc π (Definition 1, Axioms 2+3), $1/4$: inverse of the ball value 4 (Definition 2)	Only registered numbers of Paper I [5], Table 7

Note on notation: the corresponding card of the complete system [6] records the 8 in a double reading – the 8 bits of the d-ring ([6] Axiom 15) and the read cost 8 (Definition 1); the present paper uses the latter reading, the one within the 4-axiom scope.

The numerical comparison in detail [6]: the derived value is $\sqrt{3} \ln 3/8 = 0.237857$; the modern LQG literature value is the transcendental-equation root of Domagała-Lewandowski and Meissner, 0.237533 [20,21]; the deviation is 0.136%. The literature value is an exact root with no error bar, so this deviation is not a statistical agreement but a definite discrepancy, and until its origin is accounted for the status of this approach is a hypothesis. The old ABCK value of the literature, $\ln 2/(\pi\sqrt{3}) = 0.12738$, misses the modern value by 46%; the runner-up candidates tried in the course of this derivation, $3/(4\pi)$ (0.51%) and $3 \ln 2/(2\pi\sqrt{2})$ (1.48%), also fall short of $\sqrt{3} \ln 3/8$ (the attempt record is in the corresponding card of [6]).

The area-gap derivation in steps [6]: the minimal eigenvalue of the LQG area spectrum is $\Delta A = 8\pi\gamma\sqrt{j(j+1)}l_p^2$, and at the minimal representation $j = 1/2$, $\sqrt{j(j+1)} = \sqrt{3}/2$, so $\Delta A = 4\pi\sqrt{3}\gamma l_p^2$. Substituting $\gamma = \sqrt{3} \ln 3/8$, the $\sqrt{3}$ of the spin Casimir multiplies the $\sqrt{3}$ of the 3-axis norm into the integer 3, and the closed form with no surviving irrational root, $\Delta A = (3\pi/2) \ln 3 l_p^2 = 5.177 l_p^2$, holds. The deviation from the literature value $4\pi\sqrt{3}\gamma_{\text{lit}} l_p^2 = 5.170 l_p^2$ is the 0.136% inherited from γ ; this item is subordinate to the γ derivation and falls with it if γ is rejected.

The QNM correspondence in detail [6]: the asymptotic QNM damping spacing $M \Delta\omega_I = 1/4$ is the exact value fixed by Motl's analytic proof [22], so its coincidence with the inverse of the ball value 4 carries zero error. The real part $M \omega_R = \ln 3/(8\pi) = 0.043712$ agrees with Nollert's numerics [23], 0.043712. The area quantum $\Delta A = 4 \ln 3$ that Hod [16] obtained by Bohr correspondence decomposes as the product of the domain count 4 and the 3-state information $\ln 3$. Furthermore, taking the ratio of the LQG area gap to the Hod quantum, $\frac{(3\pi/2) \ln 3}{4 \ln 3} = \frac{3\pi}{8}$, cancels $\ln 3$, and the long-standing tension between the two camps (LQG vs. the QNM heuristic) over the area quantum reduces to the difference of a single geometric factor.

In LQG, γ is a free parameter the theory does not determine. That the registered-number combination $\sqrt{3} \ln 3/8$ of the present axiomatic system approaches its literature value within 0.136% is a structural coincidence of the same shape as the Hamming coincidence of Paper I – not a quantitative derivation but an a-posteriori match in which the same number combination appears in an independent discipline – and its status is confined to axiom-described (III). The origin of the residual 0.136%

deviation (the infinite spin sum of the transcendental equation vs. the finite truncation of the ring cycle) is recorded as an open item in [6], and the precise derivation is deferred to a later paper.

7.2. Forward-Testable Item: The Hubble Tension

The constituent ground recorded by the corresponding card of the complete system [6] is observer-occupied state count: the observer occupies the 7 CAS degrees of freedom ([6] Axiom 11), and subtracting the 7 occupied by the observer from the total state count 57 leaves the effective count $50 = 57 - 7$. The CMB inversion reads the observer-inclusive whole (57) while the local distance-ladder measurement reads the observer-exclusive effective count (50), so one and the same expansion rate is observed differing by the ratio $\sqrt{57/50}$ – the correction factor is the ratio of “whole including observer” to “effective excluding observer.” Numerical result and comparison:

$$H_0^{\text{local}} = H_0^{\text{CMB}} \times \sqrt{\frac{57}{50}} = 67.36 \times 1.06771 = 71.92 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (6)$$

Planck 2018 [18]: $H_0^{\text{CMB}} = 67.36 \pm 0.54$. SH0ES 2022 [19]: $H_0^{\text{local}} = 73.04 \pm 1.04$. Propagating the Planck error into the prediction gives 71.92 ± 0.58 ; the error combined with the SH0ES error is 1.19, so the difference from the measurement is 0.9σ . The original chain of the complete system [6] takes its own derived value $H_0^{\text{CAS}} = 67.92$ as input, yields 72.52, and judges 0.71% against SH0ES; the present paper conservatively recomputed with the Planck measurement as input. This item has the same status as the η_B of Paper I – a forward item testable/falsifiable by future refinement (ladder cross-checks, CMB-S4) – and the correspondence is falsified if the resolved tension keeps departing from the ratio 1.06771. Remaining tasks recorded in [6]: proving the necessity of the square-root form of the correction factor, and checking whether the same observer effect induces analogous corrections in other cosmological quantities (Ω_m, σ_8).

8. Discussion

8.1. The Discrete Tier of Creation Probability and $137 = 128 + 9$

By Theorem 3 the solution of Eq. (1) is a $\delta = 0$ description. The transition to firing – the creation of the universe – is in the present system a discrete transition of the 7-bit FSM state space (state count $2^7 = 128$, Proposition 1), and the hypothesis that creation-related probabilities lock onto the $n/128$ lattice is recorded in [6]. This 128 is the same number as in the $137 = 128 + 9$ structure of Paper I [5], Section 8.1.2 (the nearest integer to $1/\alpha(M_Z) \approx 128$). The registered decomposition of the complete system [6] is $137 = 8 + 128 + 1$ (machine 8, states 128, equals sign 1), while Paper I’s $128 + 9$ is the category combination of structure 128 with residual cost 9 – a different bundling of the same number (Paper I [5], Section 8.1.2) – so the low-energy-limit electromagnetic constant and the creation tier share one and the same FSM state count. The inflationary e-folding $N_e = 57 + 3 = 60$ (cosmological-constant exponent 57 + CAS step count 3) lies on the same chain [6]. The tier distinction of this hypothesis is recorded in [6] together with a failure record: equating $1/128$ directly with the semiclassical creation probability $P = e^{-|S_E|}$ gives $|S_E| = \ln 128 \approx 4.85$, clashing by about 8 orders of magnitude with what the Planck tensor-ratio bound requires, so that equation was discarded, and $1/128$ holds only at the separate tier of a discrete state-count probability, not an exponential one. As a collateral recheck on the same chain, [6] records that the spectral index $n_s = 1 - 2/57 = 55/57 = 0.96491$ agrees with the Planck measurement 0.9649 ± 0.0042 . These are axiom-described (III) and are not used in the quantitative results of the present paper.

8.2. The Position of the No-Boundary and Tunneling Proposals

The Hartle-Hawking no-boundary wave function [7] reads in the present system as the OPERATOR-side description of the $\delta = 0$ state – the formalization of the fact that before the first tick of system time ($T_{\text{sys}} = 1$, the Big Bang) there is no timestamp record. “No boundary” is not an

assumption but the translation of “no record yet.” The observation that the sign contest with Vilenkin tunneling [8] corresponds to the two reading directions, cost/norm, and the observation that the constituent numbers (3, 8) of the no-boundary exponent $3/(8G^2V)$ overlap with axiom-registered numbers (space writes 3, reads 8), are recorded at hypothesis level in [6], and until the promotion condition (an axiomatic proof of the exponent’s necessity) is discharged they are not claimed as results of the present paper.

The signature-ladder hypothesis – (5,2), (5,1), (5,0). Pushing the two elimination modes of Section 4.3 (the in-principle exclusion of observer, the in-execution elimination of superposition) to the end erects a three-rung ladder: the whole that no one observes is (5,2); the world an observer can see (excluding only itself) is (5,1); and the screen on which observation is complete and even the superposition is recorded as determinate is (5,0), with no negative axis – the Euclidean sign. The intuition that “the recorded past is static” is thereby expressed as a signature, and a candidate seat arises here for why the no-boundary proposal draws the beginning of the universe as a Euclidean 4-geometry and why the Wick rotation (the standard technique of computing in Euclidean signature via imaginary time) works – Euclideanization read not as a computational trick but as “moving to the signature of the determinate record” (Figure 6). This ladder is first proposed in the present paper and has been registered as a hypothesis ([6] H-990); its promotion condition is establishing the formal correspondence between the (5,0) regime and the Euclidean path-integral regime.

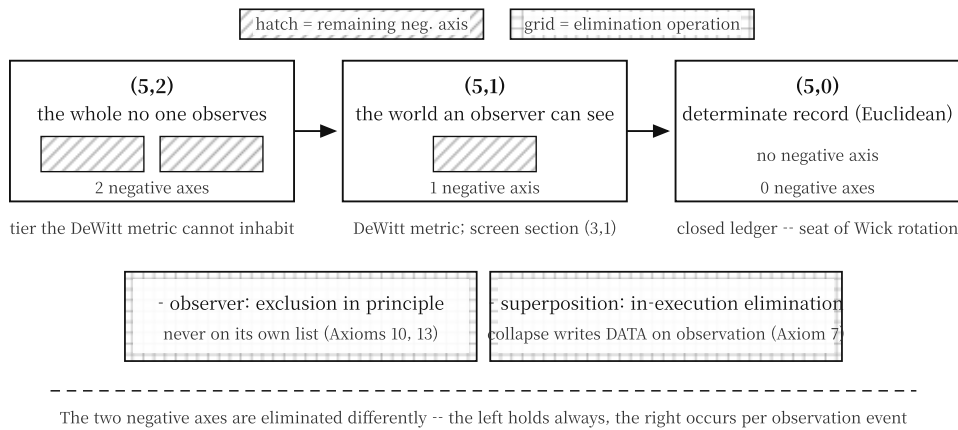


Figure 6. The signature-ladder hypothesis ([6] H-990) – the in-principle exclusion of observer lowers (5,2) to (5,1), and the in-execution elimination (collapse) of superposition lowers (5,1) to (5,0). Hatched cells are the remaining negative axes.

8.3. Contact with the Canonical Formalism

The points at which the structure of the ADM formalism [12] overlaps with the registered numbers of the present system – the 4 lapse-shift multipliers and the ball value $4 = 1 + 3$ (1 time write + 3 space writes, Definition 2), the $1 + 3$ split of the Hamiltonian/momentum constraints, and reparametrization invariance ($H = 0$) with the non-scheduling of firing (that polling runs every tick is [6] Axiom 8; that firing is not scheduled is [6] Axiom 15 proposition) – are in [6] as axiom-described (III). So is the observation that the minisuperspace truncation (cutting the degrees of freedom to (a, ϕ)) corresponds to one-axis-per-bracket truncation. The observation that in the WKB semiclassical expansion time is recovered with logarithmic structure reads as a consequence of the logarithmic compression of domain time, $t_{\text{dom}} = \log(T_{\text{sys}})$ ([6] Axiom 2 proposition and Axiom 15 proposition), and grounds unique Prediction 2 (Table 11).

8.4. Comparison of Schools

Table 8. Approaches to the problem of time – the position of the present paper.

Approach		Status of time	Position in the present system
Internal-time (Kuchar [3])	school	Designates some variable as time	Re-selection of a domain axis – movement within the slice, not recovery of the assigning axis (system time)
Page-Wootters [9]		Recovery via conditional probability	Observer ignition (1,1,0,1) – Section 5 (i). Valid, but a special case of multiple projection
Thermal (Connes-Rovelli [11])	time	The state defines the flow	Corresponds to the combination polling + RLU decay – axiom-described (III)
Thermodynamic (Kiefer-Zeh [10])	arrow	Entropy direction	Time ignition – Section 5 (ii). The RLU eviction direction
Present paper		System/domain two-tier. Elimination is projection	The level-set-projection school – the assigning axis persists as system time outside the equation

8.5. Restriction of the Search Space

The only numbers used in the class-I derivations of the present paper are the 10 admissible numbers of Paper I [5], Section 8.2 (5 structural $\{1, 2, 3, 4, 7\} + 5$ cost $\{1, 4, 5, 9, 13\}$, by the filtering rules of Paper I) and their registered identities ($7 = T(3) + 1$, $(5, 2)$, ball value $4 = 1 + 3$). The computation comparison of the component count (Table 4) is a working example of this filter: 6 itself is not a registered number (barred by its 3×2 factorization) and must be reached only by operations on registered numbers, and among the reachable paths the only one that also explains the signature is $T(3) = 7 - 1$. The auxiliary results (III) additionally use registered numbers of the complete system (8, 57, 50, 128, etc.), with each source stated in the corresponding section – these do not enter the class-I results. Moreover, what the exclusion of fitting means is the absence of continuous parameters; closing also the grammar of composite operations applied to the admissible numbers is a later task. The response to the circularity criticism that the theorems follow from the axioms, and the stance on the meta-question of axiom choice, are identical to Paper I [5], Section 8.2 and are not repeated.

8.6. Scope and Limitations

We collect the limitations stated throughout. (III) The three LQG items (Section 7.1) are structural coincidences; precise derivation is deferred. (III) The constituent-number correspondence of the no-boundary exponent (Section 8.2) is a hypothesis – no promotion before a necessity proof. (III) The derivation details of the Hubble ratio $\sqrt{57/50}$ belong to the complete system [6]; the present paper presents it only as a forward item. (IV) The formal derivation of the Born rule requires the full 15-axiom system (including the 11 axioms not reproduced here), hence out of scope [6]. (IV) The functional analysis of full superspace (measure-theoretic construction) is not within the scope of the present paper.

The three diagnoses of Section 6.3 are at hypothesis grade in the complete system [6], each with an explicit promotion condition: for the negative-probability diagnosis, an equation-level equivalence proof between Page-Wootters conditional probability and tally frequencies; for the non-normalizability diagnosis, a quantitative formalization of normalization recovery under observer conditioning; for the DeWitt-boundary-condition correspondence, fixing the scale correspondence between the scale factor a and the ring shift distance ℓ , and deriving the discrete spectrum of WDW solutions under the domain restriction $\ell \geq 1$. The promotion condition of the no-boundary-exponent correspondence (Section 8.2) additionally includes fixing the cost-dimension translation of G^2V (on which ring scale V is read). The derivation of why the DeWitt parameter $\lambda = 1$ is forced by the axioms is also incomplete. Whenever a condition is discharged the item is promoted from III to II; where it fails, the item remains a hypothesis

– neither outcome affects the class-I results of the present paper (Theorems 1-3, the derivations of Section 4).

8.7. Classification of All Results

All results of the present paper, classified by the four-level status of the Introduction.

Table 9. All results of the present paper – what 4 axioms + 1 proposition produce.

Class	Result	Remarks
I	Uniqueness of the bit reading $(0, 1, 0, 1)$	Theorem 1, under the correspondence rule of Definition 3. Exhaustive 16-combination exclusion (Appendix A)
I	Time elimination = level-set projection	Theorem 2. The now-fixing reading is declared as a premise registered in [6]
I	Uniqueness of the $\delta = 0$ translation of $\hat{H}\Psi = 0$	Theorem 3. Positive-definiteness at the two-bracket tier
I	Self-referential $+1 = \text{observer}$	Lemma 2. From the loop-path uniqueness of [6] Axiom 10
I	$\dim G_{ijkl} = T(3) = 7 - 1 = 6$	Section 4.2. Proposition 1 + Lemma 2 + Theorem 1
I	$(5, 1) = (5, 2) - \text{observer}$	Section 4.3, Route 2 (cross-derivation). Paper I Theorem 1 (cost basis) + Theorem 1
I	The formal consequence that exactly 2 switched-off axes are ignitable	Section 5. The placement of the literature workarounds into the two families is an interpretive mapping (III, Table 8)
I	Time-route derivation of $(5, 1)$ – the now-cut of the domain time	Section 4.3, Route 1 (main derivation). Theorem 1 (time = 0) and Theorem 2 combined on the premise of the structural-view $(5, 2)$ statement of the [6] Axiom 15 precinct
II	Complementarity of the two cuts – Minkowski $(3, 1)$ vs. DeWitt $(5, 1)$	Table 5. Two times, hence two places to cut. The $(3, 1)$ subframe is a registered statement of [6]
II	Necessitation of the internal-time convention for the scale factor a	Section 4.4. The remaining negative axis = trace of system time, so a playing time is consequence, not convention
II	Remaining negative axis = superposition	Formal consequence of Section 4.3 – the remainder of observer excision among the 2 reversible axes
II	Non-normalizability = absence of measure	Section 6.3. The in-system consequence is II; the quantitative formalization of recovery under observer conditioning is a promotion condition of Section 8.6 (III)
II	Consistency of DeWitt's boundary condition $\Psi = 0$ with the unreachability of $\ell = 0$	Section 6.3. The axis arithmetic is II; the quantification of the physical correspondence is a promotion condition of Section 8.6 (III)

Table 10. All results of the present paper, continued – classes III and IV.

Class	Result	Remarks
III	The observer's self-exclusion – the standing signature of the observable world = (5,1)	The physical meaning of Section 4.3, Route 2. From [6] Axioms 10, 13 and the filter-non-recording statement of Paper I [5] Theorem 1 (B)
III	The signature ladder (5,2)-(5,1)-(5,0) – candidate origin of Euclideanization	Section 8.2. Proposed here; registered in [6] as H-990. Promotion condition: formal correspondence with the Euclidean path-integral regime
III	Conformal-mode correspondence of the negative axis	Section 4.4. Interpretive proposal on top of the axis assignment (II)
III	Negative probability = error message of observer absence	Section 6.3. Non-negativity restoration under observer ignition is consistent
III	No-boundary wave function = $\delta = 0$ description	Section 8.2. The exponent-constituent correspondence is quarantined as hypothesis
III	No-boundary/tunneling sign contest = two reading directions	Section 8.2. Registered hypothesis of [6]
III	Barbero-Immirzi $\gamma = \sqrt{3} \ln 3/8$ (0.136%)	Section 7.1. Structural coincidence; precise derivation deferred
III	LQG area gap, QNM pair	Section 7.1. Downstream of γ and registered-number combinations
III	Hubble tension $\sqrt{57/50}$ (0.9σ in combined error)	Section 7.2. Forward-testable
III	Creation $n/128$ lattice, $N_e = 57 + 3$	Section 8.1. Registered hypotheses of [6]
III	ADM mapping (4 multipliers = ball value, 1 + 3 constraints), WKB logarithmic time, thermal-time correspondence	Sections 8.3, 8.4
IV	Necessity of the constituent numbers of the no-boundary exponent $3/(8G^2V)$	Section 8.2. Marked incomplete
IV	Formal derivation of the Born rule	Requires [6] Axiom 15 in full – out of scope
IV	Functional analysis of superspace (measure-theoretic construction)	Section 8.6
IV	The rule selection problem	The open meta-question of digital physics in general (Paper I [5], Section 8.6)

Classification criteria – I: formally derived from the axioms. II: the axioms force the structure, which is then mathematically determined. III: axiom-described in full – independent formal proof deferred. IV: not within the scope of the present paper. Subtotals: I: 8, II: 5, III: 11, IV: 4.

9. Conclusions

Four axioms and one proposition have fixed the axiomatic coordinates of the Wheeler-DeWitt equation and produced the component count and the signature of the DeWitt metric. **We show that axioms describe nature.**

The causal chain of the derivation: constituent inspection $\Rightarrow$ unique bit reading (0,1,0,1) (Theorem 1, exhaustive 16-combination exclusion) $\Rightarrow$ time elimination = level-set projection (Theorem 2) $\Rightarrow$ $\hat{H}\Psi = 0$ translates uniquely as $\delta = 0$ (Theorem 3) $\Rightarrow$ $\dim = T(3) = 7 - 1 = 6$, double derivation of the signature (5,1) (time route: the now-cut of the domain time, one of the two times / cost route: (5,2) – observer), with the negative axis = superposition and the trace of system time (derived) and the conformal-mode correspondence as interpretive proposal (Section 4). The (3,1) of observable spacetime enters the same picture as the complementary case of cutting the system time (Table 5). The crux is the connection to Paper I Theorem 1: from the one signature (5,2) forced by the irreversible/reversible classification of cost, one chain passes through D_5 to reach $\alpha = 1/137.036082$ [5], and the other passes through observer excision to reach the (5,1) of the DeWitt metric. The two forward chains on the same 4 axioms share one sign structure (5,2) – that the electromagnetic coupling

constant and the metric of gravitational quantization descend from the same root is the structural conclusion of this series. Further, the –observer of the cost route reads as the universal structure of observation – an observer cannot see itself, so the signature of the observable world is always (5, 1), and a completed record descends toward the Euclidean direction (the (5, 0) ladder hypothesis, Section 8.2).

The three standard criticisms were answered item by item. (R1) The absence of time is, under the present reading, the price of level-set projection, and the assigning axis persists as system time. (R2) (5, 1) is forced from the axioms, and the conformal-mode instability is reinterpreted as the scar of observer excision. (R3) We presented a structural diagnosis binding negative probability, non-normalizability, and the boundary-condition contest as symptoms of the switched-off axes – the two key equivalence proofs are incomplete (Section 8.6), and the standard fact that conditional probabilities are restored under observer ignition is consistent with the diagnosis.

As an independent check we showed that the ignition of the two switched-off axes overlaps one-to-one with the two workaround families of the literature (Page-Wootters, the thermodynamic arrow) (Section 5); as auxiliary results we presented the 0.136% approach to the Barbero-Immirzi γ and its two downstream items (Section 7.1), and as a forward item the Hubble-tension ratio $\sqrt{57/50}$ (0.9σ against measurement in combined error) (Section 7.2). The derivation of the present paper produced 7 unique predictions (Table 11). These were not added to fit the derivation but followed automatically as deductive consequences of one and the same structure. If any one is falsified, the corresponding structure is negated.

Table 11. Unique predictions of the present paper – falsification conditions explicit.

#	Prediction (axiomatic ground)	Source	Falsification condition
1	No reversal of the thermodynamic arrow even in a recollapsing universe. The arrow hangs on system time, not the scale factor – CAS records are irreversible and RLU eviction decays geometrically ([6] Axioms 6, 12, 14)	§5(ii)	Falsified if a Kiefer-Zeh-type arrow reversal [10] is established theoretically-observationally
2	WKB semiclassical time has logarithmic structure. $t_{\text{dom}} = \log(T_{\text{sys}})$ ([6] Axiom 2 proposition and Axiom 15 proposition)	§8.3	Falsified if the time parameter of the WKB expansion is established as linear
3	Time-crystal periods lock onto the $n/128$ lattice. The discrete tier of the 128 FSM states (Proposition 1)	§8.1	Observation of a time crystal with non-integer fractional period
4	No constant modulation by inter-universe phase coherence. In [6] Axiom 11, δ is one and only the observer filters are many, so third quantization – re-promoting the universe wave function to a field – is double-counting the same δ	§5(i)	Falsified by observation of coherent modulation of fundamental constants (a null prediction shared with standard physics – low discriminating power)
5	The number of interference-capable worlds does not grow with time. In [6] Axiom 11 the multiplicity of outcomes comes not from branch proliferation but from the observer filter count N , so the world count cannot exceed N	§5(i)	Falsified if growth of branch counts gains observational support (a null prediction shared with standard physics – low discriminating power)
6	Discrete fine structure of gravitational redshift ($n/128$). The system/domain two-tier time (Axiom 1)	§2.2	Falsified by confirmation of continuous redshift at arbitrary precision
7	Hubble-tension ratio $= \sqrt{57/50} = 1.06771$. The difference of observer-occupied state counts – CMB reads the whole 57, local reads the effective 50	§7.2	Falsified if the resolved tension keeps departing from this ratio (the necessity derivation of the square-root form is incomplete, Section 7.2)

Table 12 compares the production scope of the present paper (4 axioms + 1 proposition) with the complete axiomatic system [6] (15 axioms). The present paper has treated only the minimal axiom set needed for the DeWitt-metric derivation.

Table 12. Production scope – the present paper and the complete axiomatic system.

Item	Present paper	Complete system [6]
Number of axioms	4 + Proposition 1 + 2 registered statements of [6]	15
Axiom-derived (I)	8	–
Axiom-determined (II)	5	–
Axiom-described (III)	11	–
Results subtotal (Table 9 classes)	24 + out-of-scope 4	1,096 or more (Paper I [5], Table 14)
Unique predictions (falsification conditions explicit)	7 (Table 11)	130

Table A1 (Appendix A) records the exhaustive 16-combination exclusion of Theorem 1. In the complete axiomatic system [6], the coupling-constant hierarchy, quark masses, and the cosmological constant ($\Lambda_P^2 \sim \alpha^{57}$) have already been produced; their scholarly presentation is the task of later papers. Paper I [5] and the present paper are two forward chains on the same 4 axioms – (5,2) $\Rightarrow$ α and (5,2) $\Rightarrow$ (5,1) – and the third paper of the series will treat the precise consequences of the system/domain two-tier time structure (the relativistic time-dilation family).

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Abbreviations

The following abbreviations are used in this manuscript:

WDW	Wheeler-DeWitt
CAS	Conditional state transition (Compare-And-Swap)
R, C, S	Observe (Read), Compare, Record (Swap)
RLU	The Banya counterpart of the Least Recently Used cache-eviction rule – addressless indexing and grip lifetime management
FSM	Finite state machine
LQG	Loop quantum gravity
QNM	Quasinormal mode
ADM	Arnowitt-Deser-Misner canonical formalism

Appendix A.

Appendix A.1. The Exhaustive 16-Combination Exclusion of Theorem 1

The four verdict criteria (Theorem 1, proof, (i)-(iv)) applied to all 16 bit combinations. Notation: (observer, superposition, time, space).

Table A1. Exhaustive verdicts for the 16 domain 4-bit combinations.

Combination	Corresponding formalism	Verdict and ground
(0,0,0,0)	Empty set – no constituents	Excluded. Violates (i), (ii) – neither Ψ nor a domain of definition
(0,0,0,1)	Classical static 3-geometry	Excluded. Violates (i) – no Ψ
(0,0,1,0)	A description with only a clock	Excluded. Violates (i), (ii), (iii)
(0,0,1,1)	Classical spacetime – Einstein equations	Excluded. Violates (i), (iii) – no Ψ
(0,1,0,0)	Pure superposition without recording surface	Excluded. Violates (ii) – no domain of definition
(0,1,0,1)	Wheeler-DeWitt $\hat{H}\Psi[h_{ij}] = 0$	Adopted. All of (i)-(iv) satisfied; sole survivor
(0,1,1,0)	Time-dependent state without domain of definition	Excluded. Violates (ii), (iii)
(0,1,1,1)	Time-dependent Schrödinger / Tomonaga-Schwinger	Excluded. Violates (iii) – contradicts the elimination of t
(1,0,0,0)	A filter with no object	Excluded. Violates (i), (ii), (iv)
(1,0,0,1)	Observational description of a classical configuration	Excluded. Violates (i), (iv)
(1,0,1,0)	Classical clock observation	Excluded. Violates (i), (ii), (iii), (iv)
(1,0,1,1)	An observer on classical spacetime (relativistic observer physics)	Excluded. Violates (i), (iii), (iv)
(1,1,0,0)	Measurement formalism without domain of definition	Excluded. Violates (ii), (iv)
(1,1,0,1)	Page-Wootters conditional-state formalism [9]	Excluded. Violates (iv) – contradicts the absence of an observer. The ignition direction of Section 5 (i)
(1,1,1,0)	Observation-superposition-time description without recording surface	Excluded. Violates (ii), (iii), (iv)
(1,1,1,1)	Fully fired screen physics (the effective description of standard quantum mechanics)	Excluded. Violates (iii), (iv) – contradicts the $\delta = 0$ description

Of the 16 combinations, exactly one satisfies all four verdicts: (0, 1, 0, 1). Every excluded row states at least one violated verdict, and the verdicts are independent, arising from different constituents.

Appendix A.2. Explicit Numerical Evaluation of the Auxiliary Results

Step-by-step numerical evaluation of the two items of Section 7:

$$\gamma = \frac{\sqrt{3} \ln 3}{8} = \frac{1.732051 \times 1.098612}{8} = \frac{1.902852}{8} = 0.237857. \quad \gamma_{\text{lit}} = 0.237533 [20, 21], \quad \frac{|\Delta\gamma|}{\gamma_{\text{lit}}} = 0.136\%.$$

$$\sqrt{\frac{57}{50}} = \sqrt{1.14} = 1.067708. \quad H_0^{\text{local}} = 67.36 \times 1.067708 = 71.92. \quad \sigma_{\text{pred}} = 71.92 \times \frac{0.54}{67.36} = 0.58.$$

$$\frac{73.04 - 71.92}{\sqrt{1.04^2 + 0.58^2}} = \frac{1.12}{1.19} = 0.94\sigma \approx 0.9\sigma [18, 19].$$

$$\Delta A = \frac{3\pi}{2} \ln 3 = 4.712389 \times 1.098612 = 5.177 l_P^2. \quad M \omega_R = \frac{\ln 3}{8\pi} = \frac{1.098612}{25.132741} = 0.043712. \quad M \Delta \omega_I = \frac{1}{4} = 0.25.$$

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